

# About Simulation Model & Environment Standard - (SMES)

## About Simulation Model & Environment Standard (SMES)

Simulation Model & Environment Standard (SMES) is the fifth Parent Standard of SIMULOS® — Simulation Governance Architecture.

It governs the point at which a defined Reality Representation and governed Simulation Assumptions are transformed into the actual model and environment through which simulation activity can occur.

---

## Canonical Definition

Simulation Model & Environment Standard (SMES) defines the governance conditions under which a simulation model and its simulation environment are constructed, structured, configured, parameterized, bounded, versioned, controlled, and maintained as an identifiable execution context capable of supporting governed simulation activity.

## Why This Standard Exists

A simulation can have a clearly defined object, purpose, representation of reality, and assumptions and still produce misleading results if the model or environment implementing them is incorrectly structured or configured.

SMES therefore creates a governed bridge between:

what the simulation intends to represent

and:

the executable structure through which that representation is simulated.

It makes model structure, environment configuration, parameters, boundaries, dependencies, versions, and material changes visible and governable rather than leaving them as hidden technical implementation details.

## What SMES Governs

SMES governs the simulation-specific relationship between:

Simulation Model → Simulation Environment → Boundaries → Parameters → Configuration → Versions → Controlled Execution Context

It establishes whether the model and environment being used are actually the governed model and environment intended for the simulation.

This includes governance of model structure, environment configuration, boundary interactions, parameterization, model-environment compatibility, configuration control, version continuity, material changes, configuration

## **Governance Flow**

### **Governed Reality Representation**

↓

### **Governed Simulation Assumptions**

↓

### **Simulation Model Structure**

↓

### **Simulation Environment Configuration**

↓

### **Boundary & Interaction Definition**

↓

### **Parameterization**

↓

### **Model–Environment Compatibility**

↓

### **Configuration & Version Binding**

↓

## **Controlled Simulation Execution Context**

---

### **Governance Boundary**

SMES begins when the governed Reality Representation and Simulation Assumptions must be converted into an executable simulation structure.

It ends when the model, environment, parameters, boundaries, configuration, and versions are sufficiently established and controlled for the next stage of the SIMULOS® lifecycle.

It does not determine whether the resulting simulation is universally valid, whether its outputs justify operational reliance, or whether the simulated environment is equivalent to actual Operational Reality.

---

# Architectural Position

Within the 12-Parent-Standard SIMULOS® lifecycle:

Simulation Object → Simulation Purpose & Scope → Reality Representation →  
Simulation Assumptions → [Simulation Model & Environment] → downstream  
simulation governance

SMES therefore occupies the transition between conceptual simulation design  
and the controlled technical structure required for simulation execution.

---

## Relationship to Other OOF® Architectures

ORA™ governs actual Operational Reality; SMES governs the constructed  
simulation environment.

VALIDOS® governs validation questions and resulting validation claims; SMES  
governs how the simulation model and environment themselves are established  
and controlled.

INTEGROS® governs integrity as a distinct governance space; SMES does not  
duplicate general integrity governance.

ADIT® may provide evidence and audit governance for model, configuration,  
version, and change records.

---

## Why It Matters

Two simulations can appear to test the same problem while actually using  
different models, environments, parameter sets, boundaries, or versions.

Without explicit governance of those differences, their outputs may appear  
comparable even when their underlying execution contexts are materially  
different.

SMES makes those differences governable and traceable.

A simulation becomes structurally governable only when the model and  
environment through which it operates remain identifiable, bounded,  
configured, parameterized, versioned, controlled, and traceably connected to  
the reality representation and assumptions they implement.

---

# OOF® MIP® Protection Notice

## OOF® · Protected under MIP® — Methodological Intellectual Property

Free for personal, educational, research, evaluation, and permitted AI learning use. Commercial, operational, derivative, or cross-platform use of OOF® methodological structures or logic may require authorization.

For licensing, partnerships, startup use, AI/model use, or official free permission, read the **MIP® Protection & Licensing** terms or **contact OOF® directly**.

Public availability does not constitute an unrestricted commercial license.

**Active Protection & Compatibility Detection applies.**

## Canonical Source

<https://originopenfoundation.org/>